# Jessie Yuan

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## Education

**Carnegie Mellon University (CMU)** – *Pittsburgh, PA* Aug. 2022 – May 2026
B.S. in Computer Science, Minor in Mathematics

- GPA: 4.0/4.0
- **Relevant Coursework:** Robot Learning (16-831), Fundamentals of Human-Robot Interaction (16-701), Machine Learning (10-701), Embodied AI Safety (16-886), Algorithm Design and Analysis (15-451)

## Research Experience

**Interactive and Trustworthy Robotics (Intent) Lab** – *Robotics Institute, CMU* May 2025 – May 2026
Undergraduate Researcher, *Advised by Prof. Andrea Bajcsy*

- Implemented system using world models and VLMs to translate rollouts from a diffusion policy into textual narrations and select appropriate robot behaviors based on nuanced language instructions.
- Applied conformal prediction to quantify VLM uncertainty, distinguishing between scenarios where multiple valid behaviors exist versus when the base policy cannot accomplish the task.
- Tested various methods to improve confidence score quality, ultimately implementing marginalization over user intent to explicitly model ambiguity in human commands.
- Developed a residual learning pipeline enabling user correction of base policy failures and iterative policy improvement from collected corrections.

**Robotic Caregiving and Human Interaction (RCHI) Lab** – *Robotics Institute, CMU* Feb. 2023 – May 2025
Undergraduate Researcher, *Advised by Prof. Zackory Erickson*

- Developed a ChatGPT-driven speech interface for a commercial feeding robot through prompt engineering.
- Designed a ROS-based system to collect, label, and process data from wearable sensors; cleaned, visualized, and performed feature extraction on this data; and trained and deployed binary classification and regression models using PyTorch.
- Designed, organized, and conducted three human studies, including procedure documentation, participant interaction, and data analysis to collect training data and evaluate system efficacy and user experience.

## Publications

**J. Yuan***, Y. Wu*, and A. Bajcsy, "When to Act, Ask, or Learn: Uncertainty-Aware Policy Steering", to appear in *Robotics: Science and Systems (RSS)*, 2026.

A. Padmanabha*, **J. Yuan***, T. Mehta, R. K. Jenamani, E. Hu, V. de León, A. Wertz, J. Gupta, B. Dodson, Y. Yan, C. Majidi, T. Bhattacharjee†, and Z. Erickson†, "WAFFLE: A Wearable Approach to Bite Timing Estimation in Robot-Assisted Feeding", in *The ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2026. **Best Systems Paper.**

A. Padmanabha*, **J. Yuan***, J. Gupta, Z. Karachiwalla, C. Majidi, H. Admoni, and Z. Erickson, "VoicePilot: Harnessing LLMs as Speech Interfaces for Physically Assistive Robots", in *The ACM Symposium on User Interface Software and Technology (UIST)*, 2024. **Oral Presentation.**

**J. Yuan**, J. Gupta, A. Padmanabha, Z. Karachiwalla, C. Majidi, H. Admoni, and Z. Erickson, "Towards an LLM-Based Speech Interface for Robot-Assisted Feeding", in *Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (UIST Adjunct)*, 2024. **Live Demonstration.**

## Honors & Awards

**Graduate Research Fellowship** 2026
National Science Foundation (NSF)

**President's Fellowship** 2026
Georgia Institute of Technology

**Best Systems Paper Award** 2026
The ACM/IEEE International Conference on Human-Robot Interaction (HRI)

**Summer Undergraduate Research Fellowship** 2023
Carnegie Mellon University

## Teaching Experience

Teaching Assistant for 21-128: Mathematical Concepts and Proofs, *CMU* Fall 2024
Teaching Assistant for 15-251: Theoretical Computer Science, *CMU* Spring 2024, Fall 2023
Teaching Assistant for 15-122: Imperative Computation, *CMU* Spring 2023